Worked solutions to selected exercises from
Richard Hammack, *Book of Proof, edition 2.2*

9 Disproof

Each of the following statements is either true or false. If a statement is true, prove it. If a statement is false, disprove it. These exercises are cumulative, covering all topics addressed in Chapters 1-9.

2. For every natural number n, the integer $2n^2 - 4n + 31$ is prime.

Disproof. The statement is false. For a counterexample let $n = 31$. Then $2n^2 - 4n + 31 = 2(31^2) - 4(31) + 31 = 1829 = 31 \cdot 59$, which is not prime. $\square$